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Hybrid standalone microgrid for agricultural last-mile: A techno-economic analysis

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Abstract

Last-mile communities require electricity for socio-economic development, but the access is limited and majorly depends on environmentally damaging fossil fuels. Therefore, strategic energy access planning is needed to advance such communities towards sustainable development Goals (SDGs). Driven by technological advancement, falling cost of installation, growing recognition of benefits, and proven track record; microgrid networks, and green energy initiatives are evolving into a full-fledged solution for the grid disconnected remote communities. Thus, this study proposes a grid-independent hybrid micro-grid alternative for a remote South African community. It comprises photovoltaic (PV) arrays and a diesel generator. Two scenarios were evaluated vis-a-vis the inclusion or non-inclusion of a diesel-fuelled electricity generator. In addition, the CO₂ emissions are calculated to determine the environmental benefit of the proposed scenarios. The modelling was performed using a commercial software called HOMER Pro. It can be concluded that decreasing fuel prices and increasing solar irradiation can lead to a reduction in the total NPC. Also, it was concluded from a sensitivity analysis that decreasing fuel price and increasing solar irradiation could reduce the total NPC. A significant cost difference was observed between the two system scenarios, with PV-Genset having the lowest LCOE (R4.01) though the cost of operation of PV system is lower (R14,859). Overall, the proposed system would provide affordable and sustainable energy access to the last mile residential communities. © 2022 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

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Keywords: Hybrid microgrid; Last mile community; Techno-economic analysis; SDGs

1. Introduction

Energy access for all is a prime mover, and it is central to the attainment of sustainable development. Therefore, its accessibility to all will tremendously facilitate the achievement of SDGs. It has been identified that there is a positive correlation between energy accessibility and advancement towards other SDGs [1]. This necessitated the

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Nomenclature

E_{nonren}	Non-renewable electricity production [kWh/year]
E_{served}	Total electrical load served [kWh/year]
f_{pv}	PV fraction [%]
H_{nonren}	Non-renewable thermal production[kWh/year]
H_{served}	Total thermal load served [kWh/year]
$P_{\text{load},t}$	Load in time step t [kWh]
P_{load}	Annual load [kWh]
R_0	Initial investment [£]
$R_{\text{element},i}$	Element lifetime [years]
$r_{\text{load},t}$	Input operating reserve as a percentage of load in the time step t [%]
$r_{\text{load},t}$	Operating reserve as a percentage of load in the current time step [%]
r_{peakload}	Input operating reserve as a percentage of annual peak load [%]
r_{peakload}	Operating reserve as a percentage of annual peak load [%]
$R_{\text{rem},i}$	Remaining life of the component [years]
R_t	Net cash flow for each component (i.e., revenues minus costs incurred) [Rand]
T	Planning horizon [years]
t	Time of the cash flow [year]
x	Discount rate [%]
$P_{\text{load},t}$	Load in time step t [kWh]

Subscript

con	Converter
ba	Battery
pv	Photovoltaic system
dg	Diesel generator
tac	Total annualized cost

development of universal electrification framework under the UN's Sustainable Development Goals (SDGs) [2]. Electricity is so important to the extent that access at a basic level could increase productivity, reduce poverty, and provide opportunities for economic development. Even though it has been suggested as the overarching reason, the current utilization, and shift towards the exploration of renewable energy is not only due to the environmental concerns and impacts. Several communities lack access to energy resources. They majorly depend on fossil fuels, even for lighting purposes. This is due to factors such as proximity, accessibility, technical skill, and social-economic factors among others. They are often left to suffer the adverse consequences of such practices. In a World Health Organisation (WHO) report, around 4 million suffered premature death attributed to household air pollution from inefficient cooking practices [3]. Household air pollution causes about 50% of the mortality due to pneumonia among children under 5 years of age, while non-communicable disease such as stroke, ischaemic heart disease, lung cancer, and chronic obstructive pulmonary diseases has been reported [3]. Most of these people are poor and live in the last mile, where access to infrastructure and electricity is either rare or completely unavailable.

On a positive note, the number of individuals deprived of access to electricity across the globe dropped from 1 billion in 2016 to a record low of 770 million in 2019 [2]. While notable progress has been recorded in global electrification, the progress has been significantly lower among the last mile communities, as 84% of the people without access to electricity live in rural locations [2]. Thus, their core electrification challenge remains. Nearly 13% of the world population living mainly in Sub-Saharan Africa (SSA) and South Asia, still lacks access to electricity. In many developing countries, last-mile connectivity to the grid or off-grid supplies remains a challenge. For instance, a survey group from the African research network Afrobarometer polled people in 34 African nations regarding access to electricity and the availability of an accessible grid [4]. Approximately four out of ten Africans

(42 percent) do not have access to power in their homes [4]. This is because they are either in areas without an electric grid or are unconnected to an existing grid. More than half of respondents in 16 nations had no electricity access [4]. Further research has suggested that the number of people living in SSA without access to electricity will increase by 2040 when the population is estimated to reach 2 billion [5]. At the current pace, less than 60% of the people in this region will have access by 2030, this is far below the target set by SDG 7.1.1 [5,6]. The lack of access in SSA has negatively affected the GDP growth leading to accelerated poverty and inequality among the populace.

The COVID-19 pandemic has further complicated the current challenges to universal electrification in the last mile due to the forecasted reduction in the pace of electrification [7,8]. Thus, it may demand a new approach and collaborative strategies to bring the universal electrification goal back on track. This accentuates a need for a paradigm shift towards powering the last mile communities through low-cost distributed renewable resources. The provision of electricity to remote communities can improve their socio-economic standing. However, connecting the last mile to the main grid could be very difficult considering the unit cost, socio-economic status, challenging terrains, and infrastructure issues [9].

Renewable energy technologies are currently the most emphasized study topics because they can play a vital part in enhancing the quality of life by satisfying the electricity demand. If integrated into a microgrid, many benefits can be harnessed with a high level of reliability. Microgrids are now emerging as an electrification solution in the commercial markets with the advancement in RE and cost reduction of the enabling technologies. They can either stand alone or be connected to the grid. Independent connection could be beneficial using green or other standalone technologies in the event of natural disasters or power outages. Microgrids are deployed to enhance the reliability and resilience of electrical grids while managing the addition of distributed clean energy resources. They can further help advance energy justice by providing electricity in the last mile areas that are not served by centralized electricity infrastructure while reducing indoor pollution, which often arises from conventional fuel. Techno-economic models are routinely used to estimate the costs and benefits of microgrids. These tools can assist decision-makers in understanding the technical and economic viability and the long-term benefits of proposed solutions [10].

Many studies have been conducted on the technoeconomic assessment of the microgrid system, a few of which include [11–15]. However, with the number of studies reported from different geospatial locations, only a few focused on the agricultural last mile, despite their significance in the sustenance of the food value chain. Elimination of hunger and attainment of food security by 2030 is the second SDG [16] proposed by the United Nations. Human population is expected to reach 9.7 billion in 2050 [16]. With this increasing population and in the absence of electricity in the agriculture last mile, rural–urban migration may become eminent with its consequence on agricultural productivity. In fact, a study by De Clercq et al. [17] reported a 3% decrease in contribution of agriculture to global gross product (GDP) in 2018. Therefore, agricultural last mile needs to be encouraged to avert the impending global hunger [18]. Tailoring the design of microgrids to a specific case study will enhance its acceptability and provide a template that can be deployed by users that fall within a similar area of interest in electricity-based activities. Based on this background, this study carries out technoeconomic analysis of grid-independent hybrid micro-grid alternatives designed and optimized for a last mile agricultural farm in South Africa. The techno-economic evaluation proposed an optimal microgrid system considering some indicators such as operating cost (OC), return on investment (ROI), Levelized cost of energy (LCOE), and net present cost (NPC). The system was simulated using HOMER Pro software, one of the decade's top optimization tools. HOMER is one of the most extensively used software packages since it does optimization and sensitivity analysis more effectively while examining various system configurations [15,19]. Section 2 of this study presents the methodology adopted in the data collection and analysis. The results obtained and their discussion is presented in Section 3, while Section 4 brings the study to a conclusion.

2. Materials and methodology

The University of Johannesburg Process Energy and Environmental Technology Station (UJ-PEETS) supports various capacity-building and policy-influencing initiatives in South Africa to grow the green economy while providing SME engineering and technology development support. In one of such engagements, UJ-PEETS carried out an energy audit in an agricultural last-mile anonymously represented as "Farm X" in Limpopo province of South Africa. The farm is an SME focusing on the rearing of pigs, chickens, and the cultivation of vegetables. The

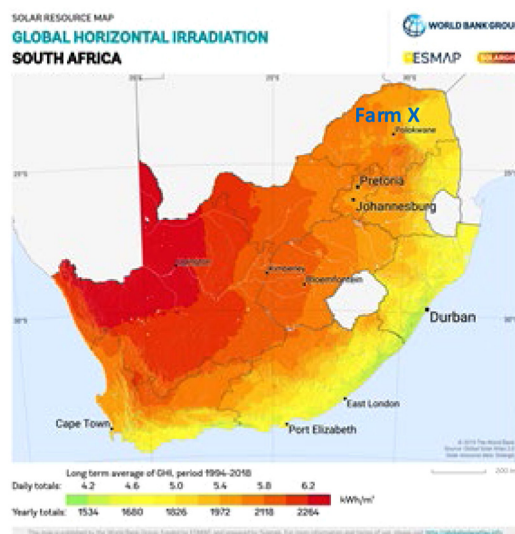


Fig. 1. Solar resource map of South Africa.

farm is not connected to the grid, though solar resources are available (see Fig. 1). The farm uses gas to heat pig pen and traditional biofuel for other activities such as cooking and lighting.

A preliminary resource assessment was undertaken using the Global Solar Atlas website and Homer Pro. This was done to establish whether the location is viable for solar energy harvesting. As shown in Fig. 2, the clearness index is estimated above 0.7 between January and April and between August and December. This means that location X has a clear and sunny condition within the timeframe. Based on the resource assessment, a hybrid power consumption mix was designed for the farm and the structure discussed in Section 2.1.

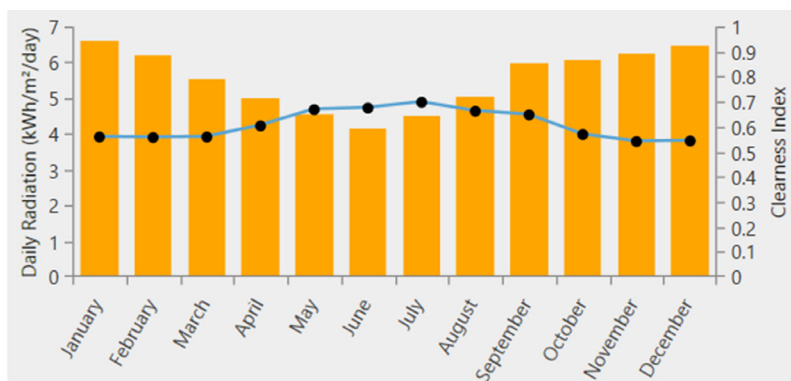


Fig. 2. Solar GHI and clearness index versus month.

2.1. Configurations of the proposed hybrid microgrid

The hybrid microgrid investigated in this study comprises photovoltaic (PV) panels integrated with battery as the storage and a diesel generator to compensate for a period of prolonged absence of sunlight. These are shown in Fig. 3 and 4. The microgrid option is chosen because there is currently no existing distribution network on Farm X. Thus, it will require significant capital expenditure to connect the farm to the grid. Although the autonomous microgrid will require capital expenditure, the benefits over the lifetime and the potential CO₂ reduction could

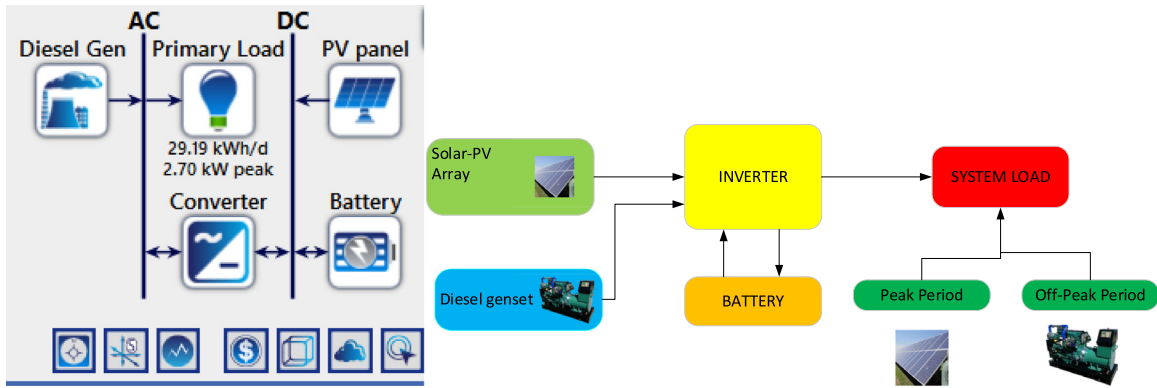


Fig. 3. Block diagrams of microgrid under first scenario.

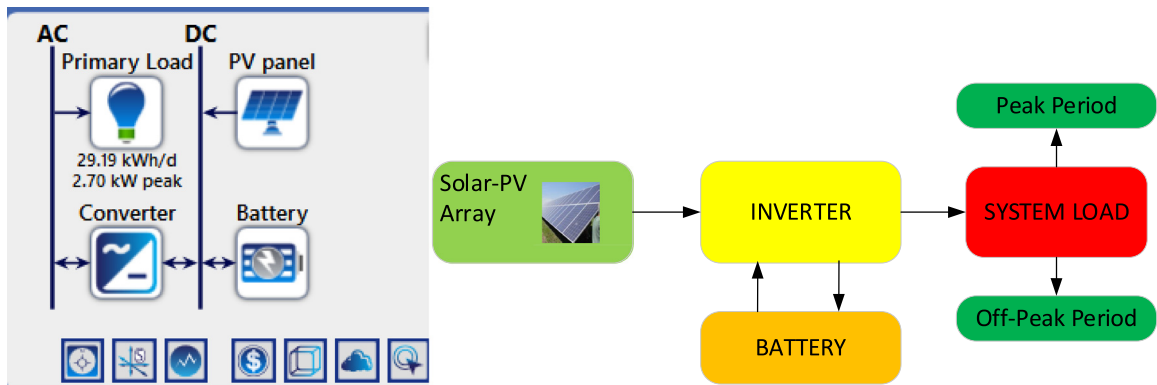


Fig. 4. Block diagrams of microgrid under second scenario.

support global decarbonization. The microgrid system was utilized to meet the load requirement of *Farm X*. To evaluate the scenario, the microgrid was extensively studied under two (2) configurations.

2.2. Mathematical problem formulation

The optimization of the microgrid in Homer Pro is designed to minimize the NPC. NPC is the combination of all the present costs incurred by the system. This also includes the installation and operation cost of all the components throughout the entire lifetime, excluding the current value of the revenue earned over the same period. In this case, the cost includes the capital, replacement, operation, maintenance, and fuel costs (due to the addition of generating set). The only revenue generated in this system is from the salvage value of the components. Mathematically, the objective function of the microgrid system cost can be expressed by Eq. (1), subject to the constraints in Eq. (2). It should be noted that $P_{shedding}$ was included for completion as the case study microgrid is not connected to the transmission grid.

$$\min (NPC_i) = \sum_i R_{0,i} + \sum_{t=0}^T \frac{R_{t,i}}{(1+x)^t} \quad (1)$$

$$\text{constrained by} \begin{cases} P_{shedding} \leq 0.05 \cdot P_{load} \\ P_{PV} \leq 0.15 \cdot E_{genset} \\ P_{load,t} \leq 0.10 \cdot P_{load,t} \\ P_{peak} \leq 0.10 \cdot P_{load} \end{cases} \quad (2)$$

Eq. (3) defined the cost across all the planning horizons for each element. The only source of revenue is from the salvage of the system as defined in Eq. (4).

$$C_{element,i} = C_{capital,i} + C_{O\&M,i} + C_{replacement,i} + C_{fuel,i} \quad (3)$$

$$C_{salvage,i} = C_{replacement,i} + \frac{R_{rem,i}}{R_{element,i}} \quad (4)$$

The capital cost (Eq. (5)) is expressed as the summation of all the capital costs due to the procurement of the solar PV, battery, converter, and diesel generator. Also, the replacement cost (C_{rep}), and operation and maintenance cost (C_{OM}) are defined by Eqs. (6) and (7), respectively.

$$C_T = C_{con} + C_{cpv} + C_{cba} + C_{cdg} \quad (5)$$

$$C_{rep} = C_{rcon} + C_{rpv} + C_{rba} + C_{rdg} \quad (6)$$

$$C_{OM} = C_{OMcon} + C_{OMpv} + C_{OMba} + C_{OMdg} \quad (7)$$

As expressed in Eq. (8), total annualized cost, C_{tac} is the summation of C_{OM} , C_{rep} , and C_T including the cost of fuel (C_{fuel}) in the entire microgrid system.

$$C_{tac} = \sum C_{element,i} = C_{capital} + C_{OM} + C_{replacement} + C_{fuel} \quad (8)$$

It is now clear that the best candidate would be the solution that gives the lower overall NPC at the project's inception while satisfying the constraints stipulated in Eq. (2). Based on the optimal solutions which Homer Pro has established, LCOE and OC were evaluated for the two optimal alternatives. This is calculated as in Eq. (9);

$$LCOE = \left(\frac{C_{tac}}{E_{served}} \right) \quad (9)$$

Further, comparative econometrics, such as Discounted payback, Simple payback, Internal rate of return (IRR), and return on investment (ROI) were reported with PV-Genset as the base system [11,20,21]. Renewable fraction is the fraction of the energy supplied to the load originating from renewable power sources. In this case, the only RE source deployed is Solar PV. Therefore, PV fraction can be calculated by Eq. (10):

$$f_{PV} = 1 - \frac{E_{nonren} + H_{nonren}}{E_{served} + H_{served}} \quad (10)$$

3. Results and discussions

It is critical to determine if Farm X's location is suitable for Solar PV microgrid. To achieve this, Homer Pro optimization software used solar resource data harnessed by NASA to determine the annual average solar irradiation. This is shown in Fig. 5. The scaled yearly average of solar irradiation is $5.52 \text{ kWh/m}^2/\text{day}$, and the scaled annual average clearness index is determined to be 0.607.

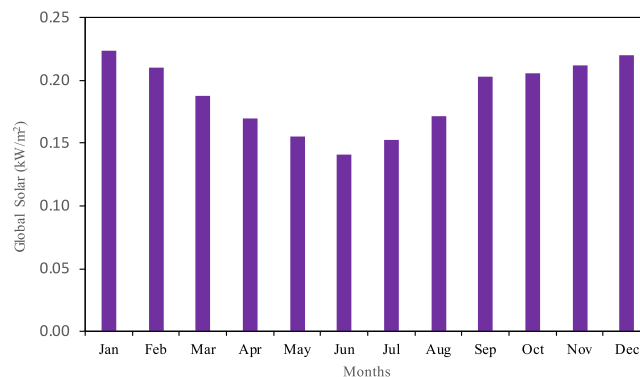


Fig. 5. Annual solar resource in Farm X.

Monthly average electrical power production of the entire energy system is shown in Fig. 6. Optimal power production was observed in the month of September. All through the year, the power production from Diesel

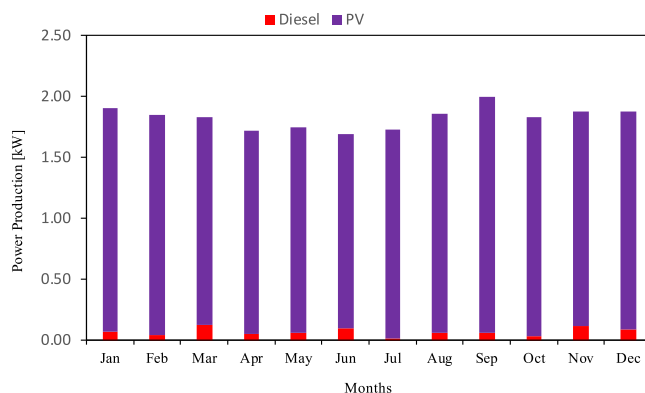


Fig. 6. Monthly average electrical outputs.

generation was minimal compared to Solar PV. Also, PV penetration appears to be generally high, with an optimal value of 95.1%. This means most of the demands could be met via solar energy. However, it could be observed that diesel Genset compensation further reduces the dependency on the solar PV system.

Fig. 7 illustrates the total electrical load served and the total RE power output. HOMER Pro has the capacity to run thousands of simulations in a matter of seconds because of its enhanced, non-derivative optimization. Therefore, in order to keep the total NPC to the barest minimum value, the unmet load should be effectively equal to zero. This means that the microgrid power system supplied all the electrical demand throughout the year. Total electrical load served was lower than the total RE production, meaning all the consumption on Farm X can be solely met through RE without needing to start up the diesel Genset except for an emergency.

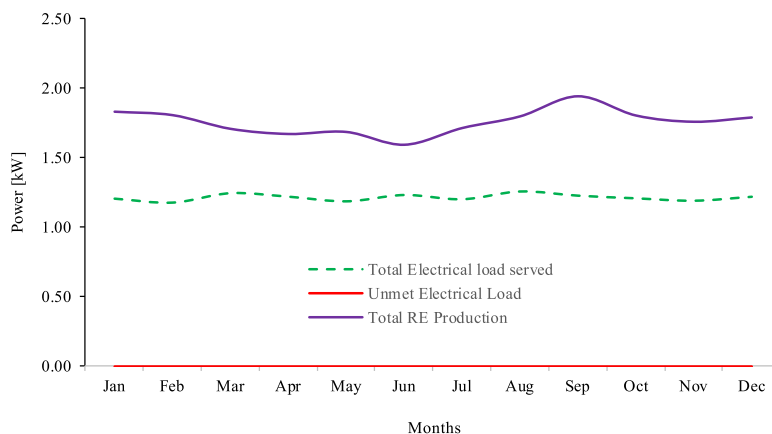


Fig. 7. Average values of total power generation, total electrical load served and unmet electrical load.

Figs. 8 and 9 show the cost components of PV and PV-Genset systems. In the case of PV systems only, it is observed that battery replacement cost is the significant cost incurred after the initial procurement. For the PV-Genset system, additional costs were incurred on the generating set's fuelling and maintenance.

3.1. Comparison of the economics

Fig. 10 shows the cost comparison between the PV-Genset (as the base case) and PV solar system only. It could take up to 8.5 years to break even after considering the present value of the future cash inflow of the project if the PV-Genset system is installed. ROI of 6.5% was recorded. This represents the yearly savings from the hybrid PV-Genset compared to the PV Solar system.

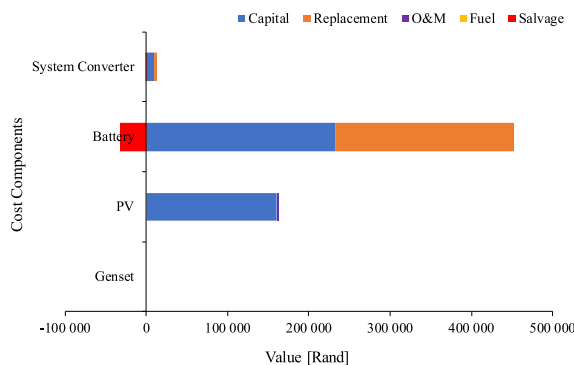


Fig. 8. Cost component for PV system.

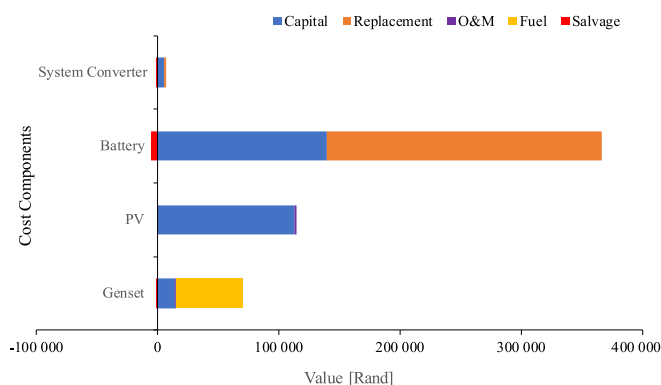


Fig. 9. Cost components of PV-Genset system.

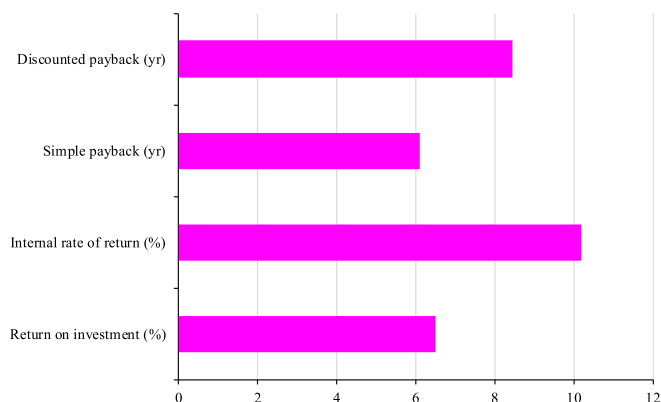


Fig. 10. Comparative economics of PV-Genset and PV system only.

Table 1 presents the cost elements of PV solar system and a hybrid PV-Genset system. Although the total NPC of the PV system is higher than PV-Genset, the operating cost is lower. Also, considering the emission reduction potential, hybrid PV-Genset exhibits the highest CO₂ emission. Thus, the PV system may be preferred considering its zero emissions, as shown in Table 2. A PV system design with PV array and battery can substantially reduce carbon emission. Taking cognisance of carbon emission in a microgrid design is of paramount importance. Moreover, this aligns the microgrid system to the global decarbonization drive [22].

Table 1. Cost elements of hybrid micro-grid.

	PV only	PV-Genset
Total NPC	593 886.80	551 766.4
LCOE	4.32	4.01
OC	14 859.36	21 607.02

Table 2. Hybrid microgrid carbon emission.

Emissions	Quantity (kg/yr)	
	PV-Diesel	Solar PV
Carbon dioxide	641	0
Carbon monoxide	4.04	0
Unburned hydrocarbons	0.176	0
Particulate matter	0.0245	0
Sulphur dioxide	1.57	0
Nitrogen oxides	3.8	0

3.2. Sensitivity analysis

A sensitivity analysis was carried out to assess the effect of diesel fuel price only on the microgrid system design, as presented in Table 3. Five prices of diesel were considered. From the table, it can be observed that as the diesel prices increase, the PV's size increases, though it was constant at diesel prices of R11.3 and R11.7, respectively. Moreover, RE fraction increases as the diesel price increases since the optimizer is working towards reducing the expenses on fuel.

Table 3. Sensitivity analysis of fuel prices.

kW/L	Diesel price (R/L)	PV array (kW)	RE fraction	NPC (R)	OC (R)
3/245	17.5	11.1	94.8	551 766	21 607
3/233	1.9	11.3	94.6	552 924	21 553
3/232	18.5	11.3	94.6	554 694	21 675
3/213	18.9	11.7	95.1	555 874	21 484
3/212	19	11.7	95.1	556 103	21 490

As shown in Fig. 11, sensitivity analysis was plotted using 3D distance method on Minitab 19. This analysis was performed to assess the combined impact of diesel and solar irradiation (these two variables determine the convectional and non-conventional generation). It can be concluded from the figure that decreasing fuel prices and increasing solar irradiation can lead to a reduction in the total NPC.

4. Conclusion

This study investigates the technoeconomic benefits of microgrids designed for the last-mile agricultural farm. The loads were analysed, and the relevant load curves were developed for 12 months. Solar resource assessment of farm X was carried out in line with NASA's databank. The optimum configuration of the microgrid was PV-Genset with the least total NPC and LCOE of R551 766.4 and R4.01, respectively, though the operating cost was observed to be higher. Also, a sensitivity analysis suggests a direct relationship between the fuel price and the PV array, such that as the price of diesel increases, the PV array size increases. Considering the need for decarbonization of energy consumption, PV system was observed to have zero carbon emission. Thus, the study suggested that the system with the lowest carbon emission should be considered to improve environmental well-being. The optimal solution obtained from the study reinforces the claim that a hybrid system is a less expensive and more environmentally friendly solution than a standalone diesel generator. In the future, the present study will be extended to consider other renewable energy solutions which may be available in agricultural last miles.

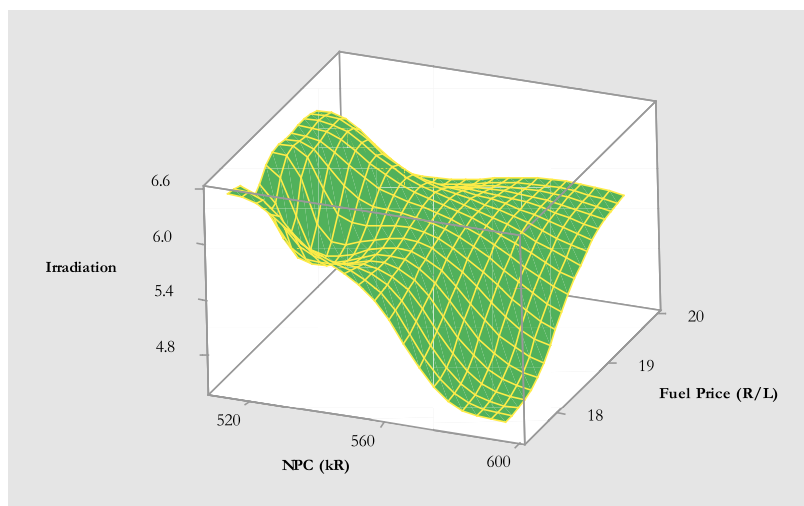


Fig. 11. Sensitivity analysis based on solar irradiation and diesel price.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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